

## Experiment No. 1

### Business Data Import and Cleaning Using Power Query

#### Aim:

To import, transform, and clean business data using Power Query in Power BI, making it analysis-ready for visualization and insights.

#### Objective:

To learn how to use Power Query in Power BI to perform data cleaning tasks such as removing unnecessary columns, handling missing values, changing data types, and applying basic transformations.

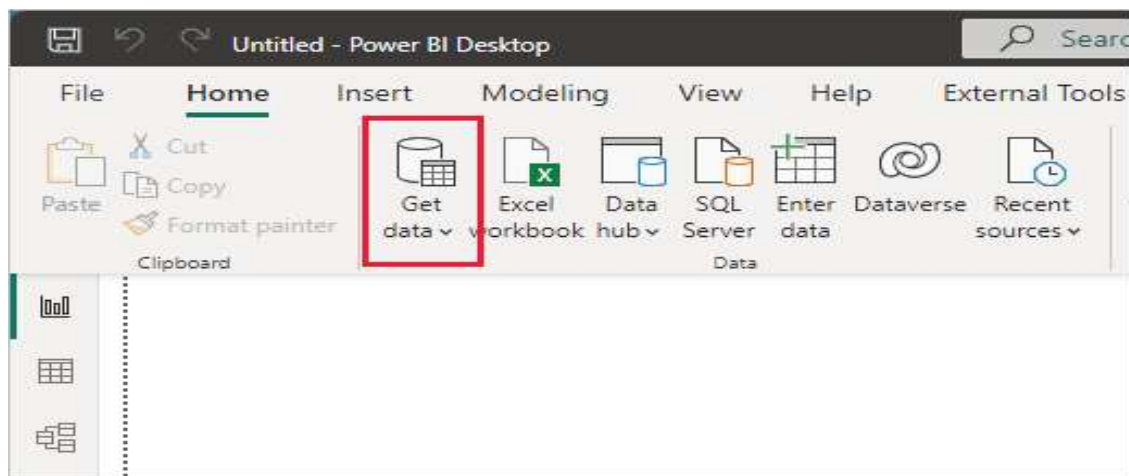
#### Input/Apparatus Used:

- Software: Power BI Desktop
- Dataset: SalesData.xlsx or CustomerData.csv (Sample business data)
- Tool Used: Power Query Editor (in Power BI Desktop)

#### Procedure:

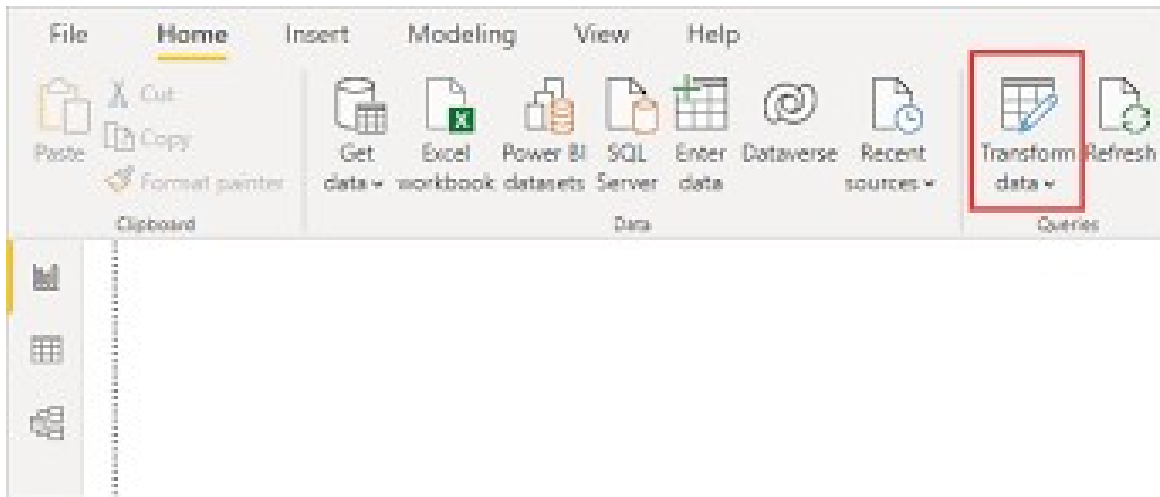
##### 1. Open Power BI Desktop

- Launch the Power BI Desktop application.
- Click on “Get Data” from the Home ribbon.



## 2. Import Business Data

- Choose **Excel** or **CSV** from the list of sources.
- Browse and select the appropriate dataset (e.g., SalesData.xlsx or CustomerData.csv).
- Preview the data and click **Transform Data** to load it into Power Query Editor.



## 3. Data Cleaning Using Power Query

Use the following transformation techniques in Power Query:

### A. Remove Unnecessary Columns

- Select columns not required for analysis and click **Remove Columns**.

### B. Rename Columns

- Double-click the column headers and assign meaningful names (e.g., change col1 to Customer Name).

### C. Change Data Types

- Ensure all columns have correct data types: Text, Number, Date, etc.

### D. Handle Missing Values

- Use options like:
  1. **Remove Rows** → Remove Blank Rows
  2. **Replace Values** → Replace null or unwanted placeholders with appropriate values.

### E. Split Columns

- If any column contains multiple values (e.g., “City – State”), use **Split Column** → By Delimiter.

#### F. Filter Records

- Use drop-down filters to remove rows with specific conditions (e.g., filtering out negative sales).

#### G. Trim and Clean Text

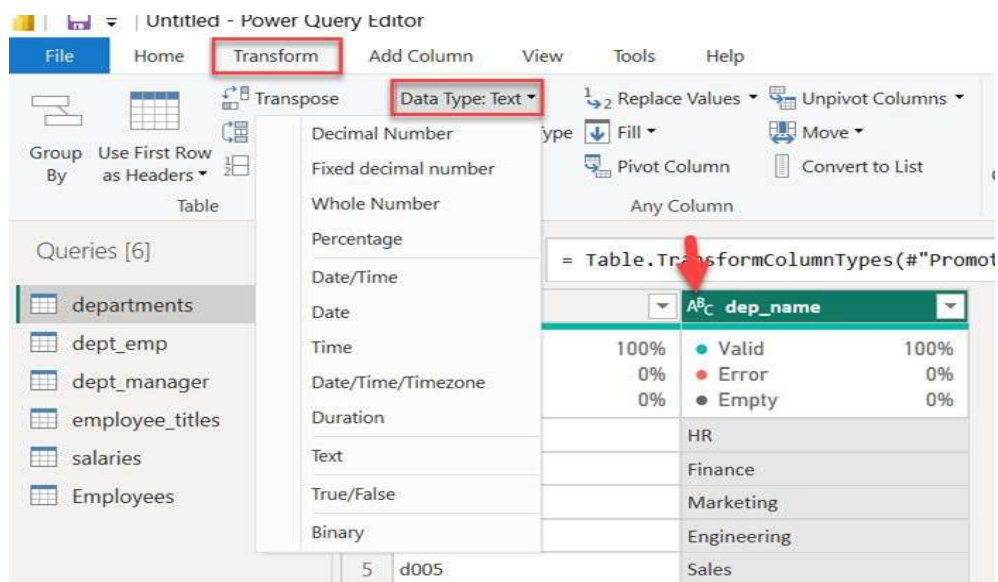
- Use **Transform** → **Format** options to remove extra spaces or unwanted characters.

#### H. Create Custom Columns (Optional)

- Use formulas to derive new fields such as profit, total price, etc.

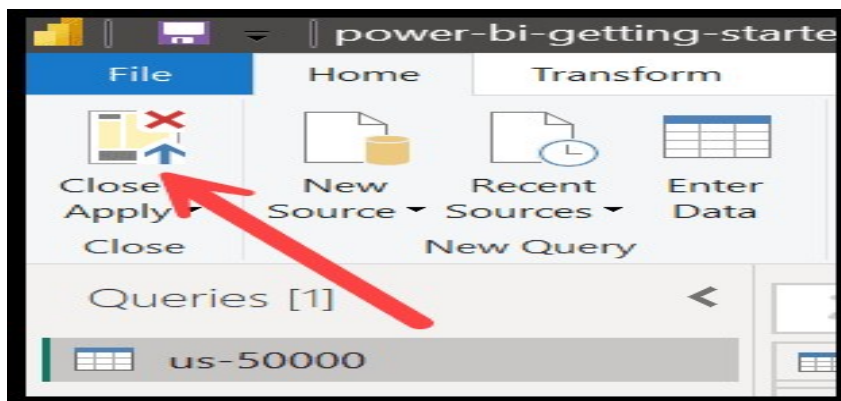
#### I. Sort Data

- Arrange records in ascending/descending order by any column.



#### 4. Apply and Load Data

- Click **Close & Apply** to save changes and load the cleaned data into Power BI for visualization.



**Parameters to Observe and Note:**

| Parameter             | Description                                              |
|-----------------------|----------------------------------------------------------|
| Source File           | Name and type of dataset used                            |
| Total Records         | Number of rows before and after cleaning                 |
| Columns Dropped       | Names or count of columns removed                        |
| Missing Values        | Strategy used to handle blanks or nulls                  |
| Data Type Corrections | List of columns where data types were corrected          |
| Transformation Steps  | Any major transformations (Split, Merge, Custom Columns) |
| Final Output Columns  | List of fields prepared for analysis                     |

**Discussion:**

Power Query significantly simplifies the data preparation process in Power BI. In this experiment, the data was cleaned using built-in transformation features such as column removal, renaming, missing value treatment, and text formatting. Proper data cleaning ensured that the dataset was reliable and ready for visualization and analytics. Using Power Query’s step-by-step approach helps users maintain transparency and reproducibility in data processing workflows.

## Experiment No. 2

### Creation of Basic Business Report in Power BI

*“Data are just summaries of thousands of stories – tell a few of those stories to help make the data meaningful.”*  
— Chip & Dan Heath

#### Aim:

To create an interactive and visually informative business report in Power BI using cleaned data, enabling users to explore key business metrics through charts, tables, and slicers.

#### Introduction

Once business data is cleaned and ready for analysis, the next step is to transform that data into meaningful visual reports. Power BI enables users to build interactive dashboards and reports that help businesses gain insights and make data-driven decisions. This experiment focuses on using cleaned data to create a basic business report in Power BI by adding visuals, formatting, and basic interactivity.

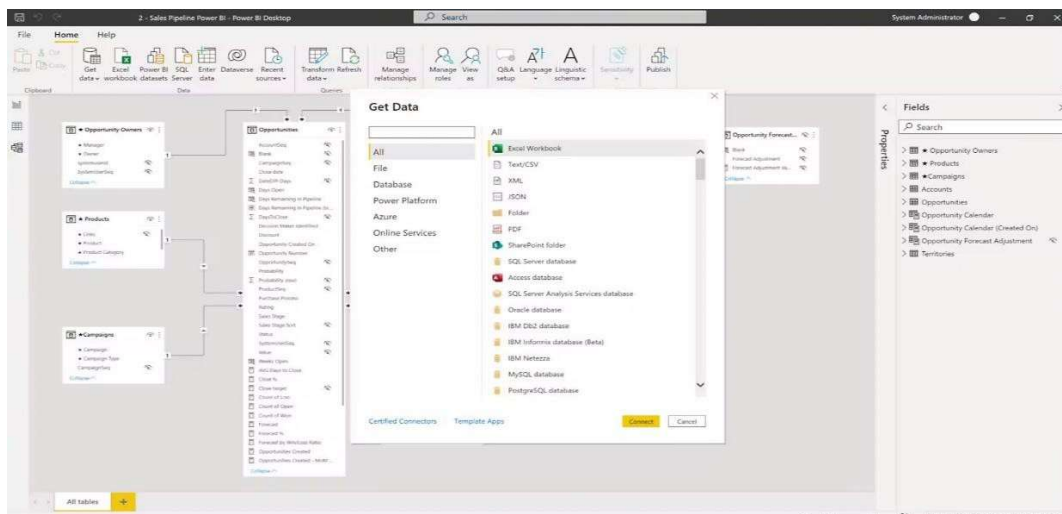
#### Objective

To design and build a basic interactive business report in Power BI using cleaned data, incorporating visual elements such as charts, tables, filters, and slicers.

#### Tasks to be Performed

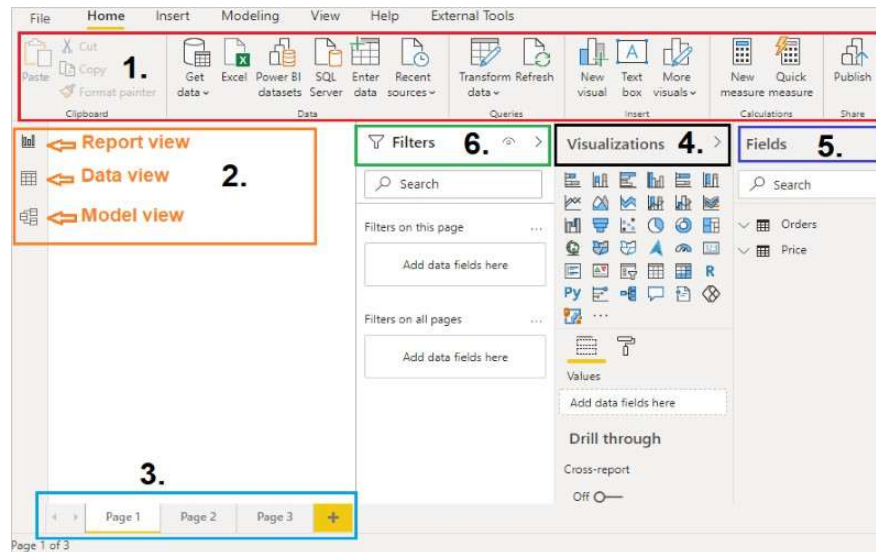
##### 1. Open Cleaned Dataset

- Launch Power BI Desktop.
- Open the Power BI file created in Experiment 1.1 or import the cleaned dataset if starting fresh



## 2. Explore the Data Fields

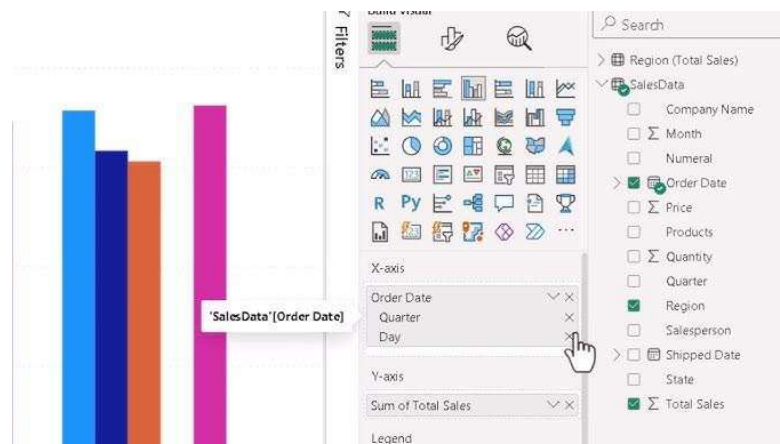
- View the **Fields** pane and understand available columns.
- Identify key metrics and dimensions for reporting (e.g., Sales Amount, Customer Name, Region, Product Category).



## 3. Create Basic Visuals

Use the **Visualizations** pane to create the following visuals:

- **Bar/Column Chart**: Display total sales by product or region.
- **Pie Chart**: Show sales distribution by category or customer segment.
- **Table/Matrix**: Present detailed data like customer-wise or product-wise transactions.
- **Card**: Show key KPIs (e.g., Total Revenue, Total Customers).



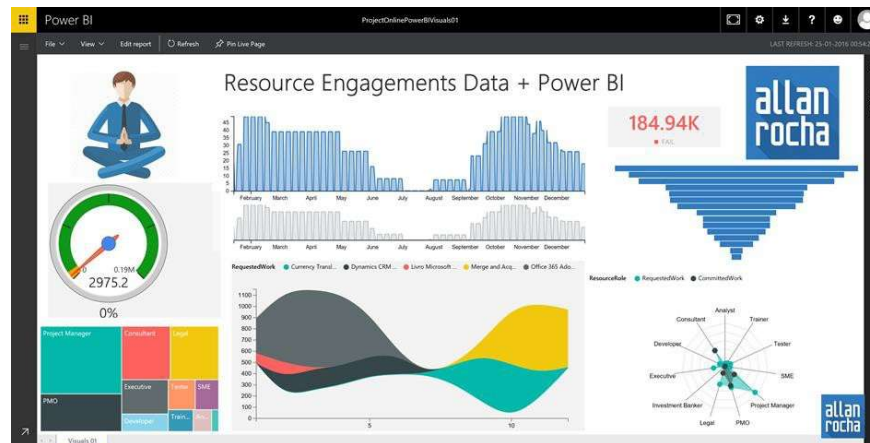
#### 4. Add Filters and Slicers

- Insert **Slicers** to enable filtering by Date, Region, Product Category, etc.
- Use **visual-level**, **page-level**, or **report-level filters** to refine data view.



#### 5. Customize Report Appearance

- Use the **Format pane** to adjust:
  - Colors, fonts, labels, and legends.
  - Data labels and titles.
  - Background and borders of visuals.



- Add **report title and company logo** using Text Box and Image.

#### 6. Add Interactivity

- Enable **cross-filtering** between visuals for dynamic interaction.
- Use **drill-down** features in charts for hierarchical data.

## 7. Save and Publish

- Save the Power BI report (.pbix).
- (Optional) Publish the report to **Power BI Service** for web access and sharing.

### Parameters to Observe and Note

| Parameter              | Description                                                                |
|------------------------|----------------------------------------------------------------------------|
| Dataset Used           | Name of the cleaned dataset/report base                                    |
| Key Metrics Visualized | KPIs such as Revenue, Customer Count, Quantity Sold                        |
| Visual Types Used      | Types of visuals included (e.g., Bar Chart, Pie Chart, Card, Matrix)       |
| Filters/Slicers Used   | Filtering tools added for interactivity                                    |
| Formatting Choices     | Colors, labels, layout, and branding decisions                             |
| Report Pages Created   | Number of report pages created and their focus                             |
| Interactive Features   | Cross-filtering, drill-down, dynamic visuals implemented                   |
| Export/Sharing Options | Whether the report was saved locally, published online, or exported to PDF |